

MARS: A Safety-Filtered Multi-Robot Area-Coverage Testbed with Classical, Random-Walk and MAPPO Controllers

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Abstract—We describe MARS (Multi-Agent Robot Swarm), an open ROS 2 Jazzy / Gazebo Harmonic testbed for cooperative area coverage by a small team of differential-drive robots. It puts three controllers behind one three-stage safety stack (two fixed override rules, a soft control-barrier-function filter, and a front safeguard): a classical decentralized planner (frontier selection, A* routing, Voronoi partitioning with consensus-based auction claims within a 3 m radio range, and a behavior tree), a random walk, and a MAPPO policy with a permutation-invariant neighbor encoder and a centralized critic, across five Gazebo worlds. In long single reference runs the classical planner reached high coverage of the discovered non-obstacle cells (85.7–100% in four worlds, 25.0% in the maze), and in an exploratory single-episode matrix a trained MAPPO checkpoint had lower observed coverage than the classical planner in all five worlds; the comparison is between an engineered system and a minimally trained reactive policy, not a matched-information benchmark. We also document three limits of the testbed: evaluation of an identical checkpoint varies between runs (a documented property that users must account for), the safety margin is soft with no formal guarantee, and the architecture’s transfer to other team sizes is untested. Code and per-run results for the reported experiments are archived with a DOI.

Index Terms—multi-robot systems, area coverage, control barrier functions, MAPPO, reproducibility

I. INTRODUCTION

Cooperative area coverage, in which a team of robots jointly sweeps an unknown environment, underlies search, inspection and monitoring. Evaluating coordination methods for it needs a simulator that models range sensing and contact, and results that others can reproduce. MARS is a testbed built for that purpose. It wraps Gazebo in a multi-agent environment, provides a classical decentralized controller and a learned MAPPO baseline behind the same safety stack, and archives per-run results.

Contributions.

- A documented testbed with five worlds, three controllers and a common safety stack, described here as implemented (Sec. III).
- Long-run reference coverage of the classical controller in the five worlds, a robot-count sweep, and a comparison with trained MAPPO baselines, all from archived runs (Sec. IV).

- An account of the testbed’s limits: non-deterministic evaluation, a soft safety margin, and claims that remain untested (Sec. V).

We make no claim that the learned policy is competitive, that the safety filter gives a formal guarantee, or that the policy transfers to other team sizes.

II. RELATED WORK

Frontier-based exploration [1] and its multi-robot extensions [2], [3] are the classical baseline for this task; the consensus-based bundle algorithm [4] supplies the auction mechanism we use to resolve duplicate target claims. MAPPO [5] is a strong baseline for cooperative multi-agent learning; we train it with RLlib [6]. The actor’s neighbor encoder is a mean-pooled embedding in the style of Deep Sets [7]; it contains no message passing, so we do not call it a graph neural network. Control barrier functions [8] give a quadratic-program safety filter, and barrier-function filters have been placed around learned policies [9]. We solve ours with OSQP [10] and add a slack variable, which makes it soft (Sec. III-C). A companion study [11] examines the MAPPO baseline in depth.

III. THE TESTBED

A. Platform and Metrics

Three TurtleBot3 Waffle robots (differential drive, 24-sector LiDAR, odometry) run in Gazebo Harmonic under ROS 2 Jazzy [12], [13]. The simulator is wrapped as a PettingZoo [14] parallel environment. The workspace is a grid of 0.4 m cells. A cell is marked visited when a robot’s pose falls in it or a LiDAR ray (up to 2.0 m) passes through it as free space. The *area-coverage rate* is

$$\text{ACR} = \frac{|\text{visited} \setminus \text{obstacle}|}{N_{\text{cells}} - |\text{obstacle}|} \times 100, \quad (1)$$

where “obstacle” cells are those the robots have registered as obstacles, so the denominator depends on what has been discovered. ACR here is therefore a *discovered-map* metric: controllers that discover obstacles at different rates have different denominators, so ACR is not a fixed-denominator measure

of physical coverage, and a ground-truth-denominator metric is not reported (Sec. V). A *wall contact* is counted when the minimum LiDAR range is below 0.14 m (the approximate sensor-to-bumper distance); an *agent contact* when two robots are closer than 0.20 m after the first 15 steps. The agent-contact threshold is smaller than the filter’s separation buffer (Sec. III-C), so a zero count does not show that the margin held.

B. Observation and Action

Each robot observes a 46-vector: 24 LiDAR sector ranges, goal distance and bearing, linear and angular velocity, and nine neighbor slots of (distance, bearing), padded with (10.0, 0.0). Actions are (v, ω) with $|v| \leq 0.22$ m/s and $|\omega| \leq 1.0$ rad/s. Because the vector has a fixed length, the network accepts up to nine neighbors.

C. Safety Stack

Every controller’s command passes through three stages. (1) *Override rules*. If the minimum LiDAR range over all beams is below 0.22 m and the commanded forward speed is positive, forward speed becomes -0.08 m/s; if another robot is within 0.45 m and within 72° of the heading, forward speed becomes -0.05 m/s. (2) *Soft CBF*. A quadratic program (OSQP) solves

$$\begin{aligned} \min_{v, \omega, \delta} & (v - v_{\text{nom}})^2 + 0.05(\omega - \omega_{\text{nom}})^2 + P\delta^2 \\ \text{s.t.} & A_k v + B_k \omega + \delta \geq -\gamma h_k, \end{aligned} \quad (2)$$

with $0 \leq \delta \leq 10$, $P = 1000$ and $\gamma = 2.0$. Barrier terms come from LiDAR returns within 0.6 m, evaluated at two lookahead points (± 0.12 m on the heading axis) with obstacle radius 0.20 m, and from neighbors within 1.2 m with a separation buffer of 0.45 m (not a physical radius). The neighbor term treats the neighbor’s velocity as bounded and unmodeled. If the QP is infeasible the filter returns $(\min(0, v_{\text{nom}}), \omega_{\text{nom}})$. (3) *Front safeguard*. Forward speed is set to $\min(0, v)$ if any of the five forward-cone beams reads below 0.24 m. The slack keeps the QP feasible when constraints conflict (for example in tight clusters), at the price of the guarantee: because of δ , the radii are soft constraints: we claim no formal guarantee and report behaviour only.

D. Classical Controller

Each robot keeps its own visited and obstacle grids, built from its LiDAR. The stack has these components:

- **Map sharing:** robots within 3.0 m merge visited grids by bitwise OR, a range-limited approximation of gossip consensus; a global grid is kept only for monitoring and for computing ACR.
- **Target selection:** frontier cells are the nearest unvisited candidates inside the robot’s own Voronoi cell relative to sensed neighbors; duplicate claims are resolved by a consensus-based auction among peers in range, and a stale claim from a peer that has left range expires after 30 steps. Unreachable targets are blacklisted.
- **Routing:** 8-connected A* on an obstacle-inflated grid.

- **Behavior tree:** in priority order, mission-complete standby or perimeter patrol, an escape maneuver ($v = -0.12$ m/s, $\omega = \pm 0.6$ rad/s) when a robot has not moved for 30 steps, frontier exploration, and a boundary-aware wander.
- **Localization:** a correlative scan matcher corrects odometry drift.

E. MAPPO Baseline

A single policy is shared by all robots. Each neighbor’s (distance, bearing) pair is embedded by a two-layer ReLU network ($2 \rightarrow 16 \rightarrow 16$), masked and mean-pooled, concatenated with the 28 ego features and passed through a two-layer Tanh actor ($44 \rightarrow 64 \rightarrow 64$). A centralized critic ($76 \rightarrow 64 \rightarrow 64 \rightarrow 1$, Tanh) takes the same 44 features plus a mean-pooled embedding ($48 \rightarrow 32 \rightarrow 32$) of the other agents’ observation-action pairs. The mean-pooled embedding is invariant to neighbor order and padding position, which `tests/test_permutation_invariance.py` checks directly. Training uses RLlib PPO [15] with a single in-process rollout worker, train batch 700, 90-step episodes, minibatch 64, 5 epochs, learning rate 10^{-4} , clip 0.2, $\gamma = 0.99$, GAE $\lambda = 0.95$ and entropy coefficient 0.01. The per-step reward is

$$\begin{aligned} r_i = & -0.01 + 5.0 \Delta d_i + 4.0 n_{\text{new}} + 20 \mathbb{I}[\text{goal}] \\ & - 8.0 \mathbb{I}[\text{wall}] - \rho_i, \end{aligned} \quad (3)$$

where Δd_i is the reduction in goal distance, n_{new} the number of cells newly visited by any robot this step, and $\rho_i = 0.5 \mathbb{I}[d_{\min, i} < 0.5 \text{ m}] + 10 \mathbb{I}[d_{\min, i} < 0.20 \text{ m}]$ with $d_{\min, i}$ the distance to the nearest other robot. The discovery term n_{new} is a team-level quantity used only in the training reward; it is not part of the decentralized observation. The goal is a random point from a fixed per-world list of safe positions, reassigned on arrival; it is less informed than the classical controller’s map-aware frontier targets. The policy is purely reactive: it has no recurrent state or local map, and no spatial-memory variant was trained. Evaluation uses greedy actions. All learned results use a single archived checkpoint (`checkpoints/mappo_baseline`, trained on the cafe); the training seed and iteration count of that checkpoint are not recorded in its archived metadata and no checkpoint-selection procedure was applied, so no across-seed estimate is claimed for it. The reward ablation below used separately trained variants.

IV. RESULTS

Protocol and caveats. Unless stated, each cell is one episode. We found that repeating an evaluation of an identical checkpoint and seed gives different outcomes (Sec. V), so single-episode numbers carry unquantified run-to-run variance; we treat them as descriptive. Raw per-run records are archived (Data availability).

TABLE I

CLASSICAL CONTROLLER, THREE ROBOTS, ONE LONG RUN PER WORLD (SINGLE-RUN REFERENCE VALUES, NOT AN ESTABLISHED BENCHMARK). CONTACTS ARE WALL / AGENT COUNTS OVER THE WHOLE RUN; CLEARANCE IS THE MINIMUM INTER-ROBOT DISTANCE.

World	Cells	Steps	ACR	Plateau	Dist.	Wall	Agent	Clr. (m)
warehouse	9,375	9,050	100.0	7,660	1263	17,018	0	0.81
office	3,500	3,500	98.2	3,466	377	0	0	0.38
depot	3,000	3,500	85.7	2,906	180	477	0	0.55
cafe	800	1,200	100.0	397	82	738	0	0.46
maze	4,800	4,000	25.0	3,961	316	1,108	0	0.26

TABLE II

EXPLORATORY SINGLE-RUN BASELINE: CLASSICAL CONTROLLER, 200 STEPS, ONE RUN PER CELL, ACR % (AGENT CONTACTS).

World	$N=2$	$N=3$	$N=5$	$N=8$
warehouse	3.4 (0)	4.8 (0)	5.3 (0)	8.7 (0)
office	5.9 (0)	6.1 (0)	12.7 (0)	46.8 (2)
depot	16.9 (0)	16.6 (0)	31.9 (0)	54.4 (0)
cafe	64.3 (0)	69.3 (0)	66.9 (2)	98.6 (8)
maze	1.1 (0)	3.2 (0)	6.0 (0)	10.0 (0)

A. Long-Run Reference Runs of the Classical Controller

Table I reports one long run per world with three robots. The step budget differs by world and was chosen to see where coverage levels off, so “steps to plateau” is the first sampled step at which ACR came within 0.5 points of its final value; it is not a controller-versus-controller comparison.

In 3 of 5 worlds (warehouse, depot, cafe) ACR reached its plateau before 90% of the run, so those budgets were long enough to see it level off; for the others the final value may still have been rising. The maze’s figure should not be read as a success or a failure by itself: we did not measure how much of its bounding box is traversable, and the ACR denominator only excludes cells where a LiDAR return has registered an obstacle. The warehouse run recorded a very large wall-contact count over its long perimeter patrol after coverage saturated, which the contact metric (sensor range below 0.14 m) makes easy to accumulate.

B. Team Size

Table II varies the number of robots from 2 to 8 in each world for 200 steps (one run per cell). Coverage generally rose with team size, though not monotonically (cafe, $N=3 \rightarrow 5$), and the 200-step horizon is far from the plateau in the larger worlds. Agent contacts occurred in 3 of 20 configurations, all at $N \geq 5$ in the two smallest arenas (cafe and office). No other explanation for that pattern was tested.

C. Learned versus Classical

Table III compares the trained MAPPO policy (trained on the cafe, zero-shot elsewhere) with the classical controller and a random walk, one 150-step episode per cell with three robots. In this single-episode matrix the classical controller had higher observed ACR than MAPPO in all five worlds. This single-episode matrix is exploratory; the companion

TABLE III

EXPLORATORY SINGLE-EPISODE BASELINE: ACR % AFTER 150 STEPS, THREE ROBOTS, ONE EPISODE PER CELL. MAPPO IS ONE CHECKPOINT (SEE TEXT).

World	Classical	Random	MAPPO
cafe	57.0	22.6	28.2
warehouse	4.3	2.6	3.3
depot	13.3	6.8	6.7
office	3.8	1.7	3.0
maze	2.7	1.5	2.2

study [11] reports a 10-episode cafe comparison in which MAPPO’s median coverage (14.5%) was below both the classical controller’s (38.6%) and a random walk’s (29.6%); the per-episode data for that comparison are not archived. Because the controllers differ in information (the classical one builds a map, MAPPO sees only local ranges and a random goal), this compares an engineered system with a minimally trained learned one, not two learned policies.

A reward ablation trained three MAPPO variants for 15 iterations with three training seeds each. Averaged over seeds, coverage was 28.0%, 31.0% and 40.5% for the baseline, a lower collision penalty and a higher discovery bonus. Paired tests across the three seeds did not reach $p < 0.05$ for either variant, so we draw no conclusion about the rewards; see the companion study for the analysis.

D. Safety Observations

Across the 20 sweep runs, the 25 matrix runs and a 27-episode ablation (72 runs), agent contacts were absent in 69 of 72, while coverage ranged from 1.1% to 98.6% (across worlds and horizons), so the contact count did not distinguish weak from strong controllers. In the office run the minimum inter-robot distance was 0.38 m, below the 0.45 m CBF separation buffer, with no contact, which is the behaviour expected of a soft constraint. Two one-off qualitative stress tests with moving hazards (head-on at 0.38 m/s and crossing at 0.22 m/s) were run once each without contact; their raw traces are not archived.

V. LIMITATIONS

- **Non-deterministic evaluation.** Three evaluations of one saved checkpoint with the same seed argument gave coverage between 24.6% and 37.8%, and contact counts of 0 to 19; the cause was not isolated: greedy actions remove policy sampling noise, and we suspect simulator or runtime timing as a contributor, but physics scheduling, message ordering and reset state were not separately tested. The three repeats used one checkpoint on the cafe for 150 steps. Single-episode cells (Tables II and III) are descriptive only.
- **Soft safety margin; no formal guarantee.** See Sec. III-C. We did not measure how often or how much the override rules and filter change each controller’s commands, so reported outcomes are those of controller plus safety stack, not of the controller alone.

- **ACR is a discovered-map metric.** Its denominator excludes only cells where a LiDAR return has registered an obstacle, so values are not directly comparable across controllers that discover obstacles at different rates, and 100% means all currently non-obstacle cells were visited, not that all physically navigable area was verified. A ground-truth-denominator ACR is future work.
- **Transfer to other team sizes is untested.** The encoder accepts up to nine neighbors and is order-invariant, but no trained policy was evaluated with a different number of robots. The sweep uses the classical controller.
- **Ablations not run.** We did not ablate the CBF, the consensus merge, the behavior tree or the scan matcher, so we cannot say how much each contributes.
- **Scope.** Simulation only, one robot model, three robots in the learned experiments, one training world, short training (about 3×10^4 steps), and no comparison with other published coverage planners.

VI. CONCLUSION

MARS provides a documented testbed with archived code and per-run results. In the reported long reference runs the classical controller’s coverage of the discovered map was high in four of five worlds, coverage generally grew with team size in the exploratory sweep, and in the single-episode matrix the observed MAPPO ACR was lower than the classical controller’s in all five worlds. These results are descriptive: evaluation is not deterministic, the learned and classical controllers differ in information and design, the safety margin is soft, and several architectural claims remain untested. We release the code, per-run logs and scripts so that others can check and extend them.

Data availability.: Code, per-run results and scripts are in the project repository, archived at <https://doi.org/10.5281/zenodo.22758286> (directory checkpoints/); GitHub: <https://github.com/harsh-pandhe/MARS>. Raw traces of the two one-off hazard scenarios (Sec. IV) are not archived.

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